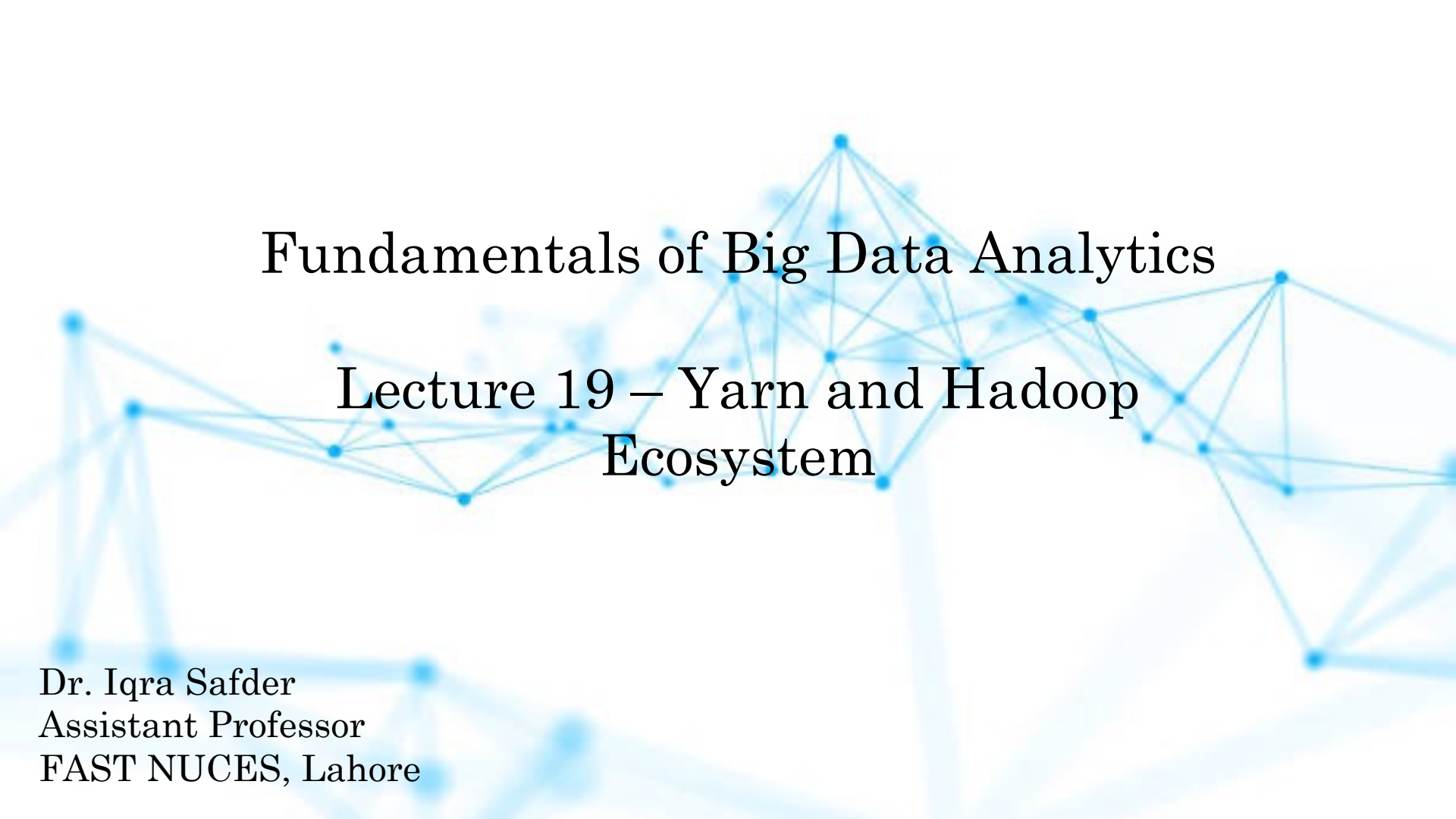
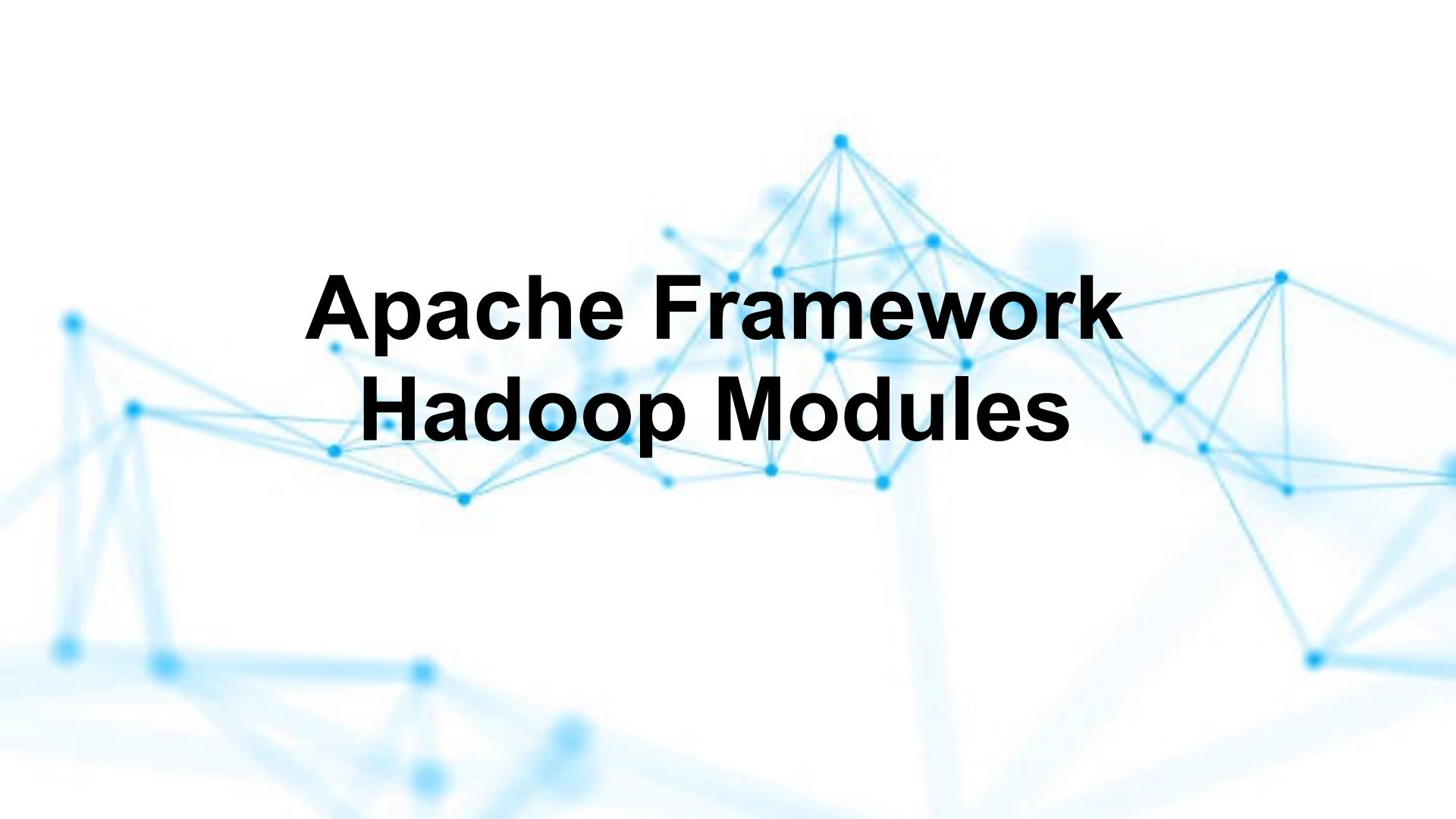


A faint, light blue background graphic consisting of a network of interconnected nodes and lines, resembling a data structure or a social network, spanning the entire slide.

Fundamentals of Big Data Analytics

Lecture 19 – Yarn and Hadoop Ecosystem

Dr. Iqra Safder
Assistant Professor
FAST NUCES, Lahore

The background of the slide features a complex, abstract network diagram. It consists of numerous small blue dots, representing nodes, which are interconnected by thin, light blue lines, representing edges. The network is dense and multi-layered, with some nodes having more connections than others, creating a web-like structure. The overall aesthetic is clean and modern, with a focus on connectivity and data flow.

Apache Framework Hadoop Modules

Apache Framework Basic Modules

Hadoop Common

**Hadoop Distributed File System
(HDFS)**

Hadoop YARN

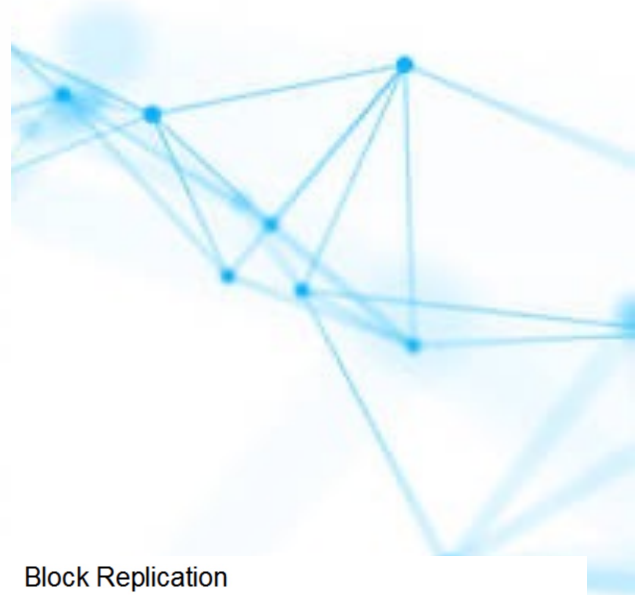
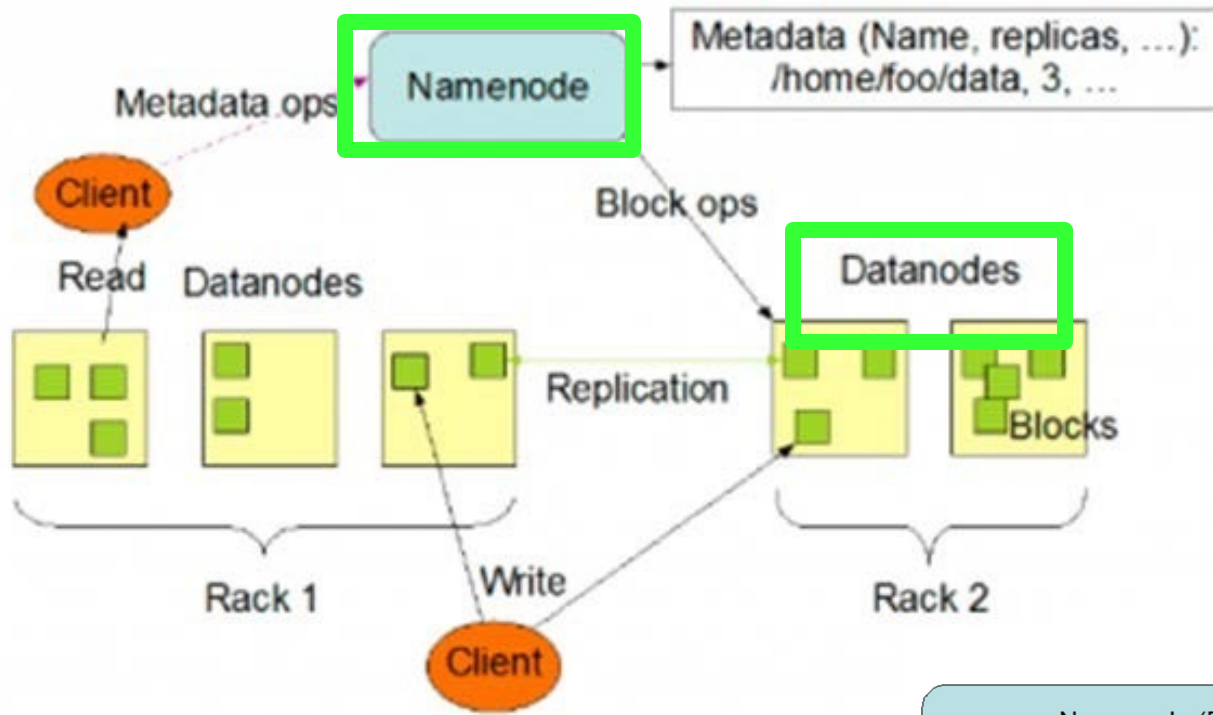
Hadoop MapReduce

HDFS

Hadoop Distributed File System

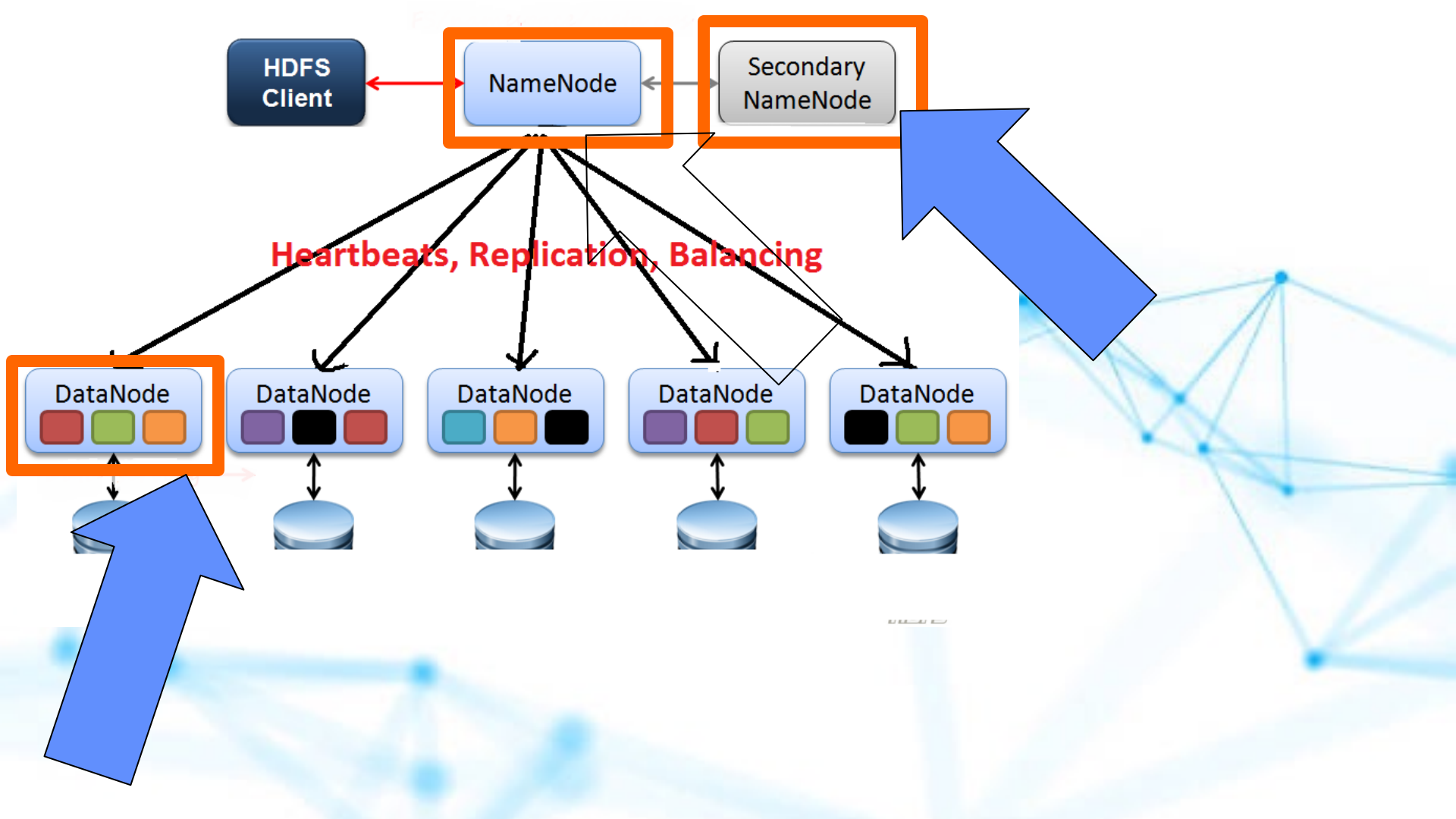
Distributed, scalable, and portable file-system written in Java for the Hadoop framework

HDFS



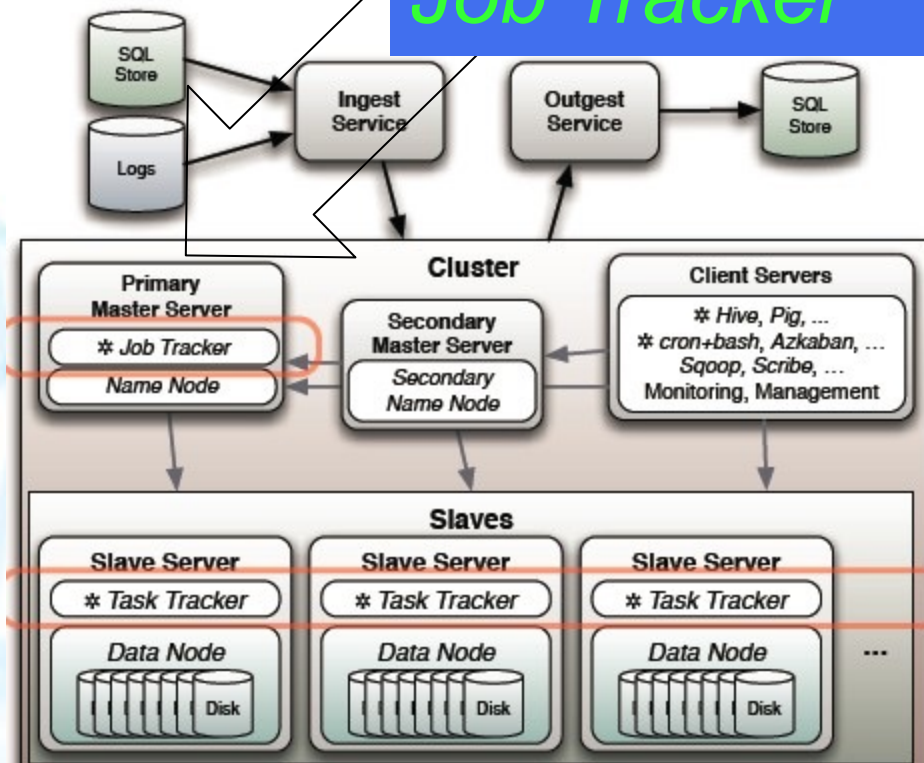
Block Replication

Namenode (Filename, numReplicas, block-ids, ...)
/users/sameerp/data/part-0, r:2, {1,3}, ...
/users/sameerp/data/part-1, r:3, {2,4,5}, ...

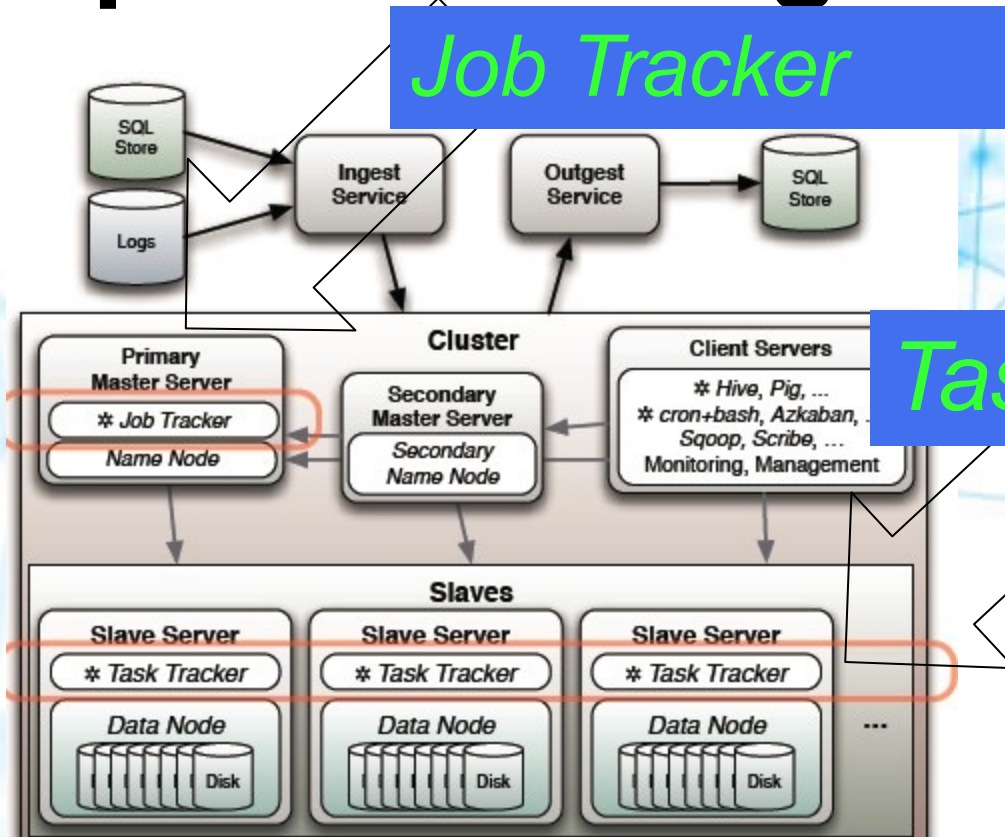


MapReduce Engine

Job Tracker

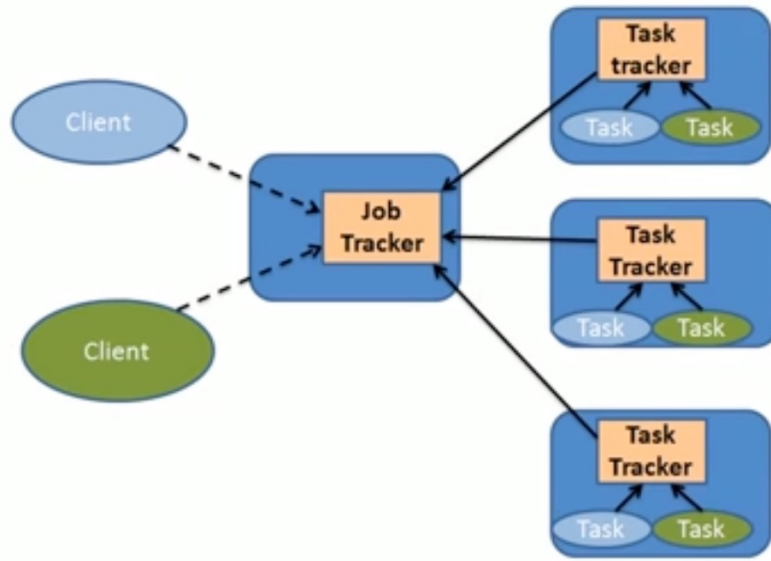


MapReduce Engine



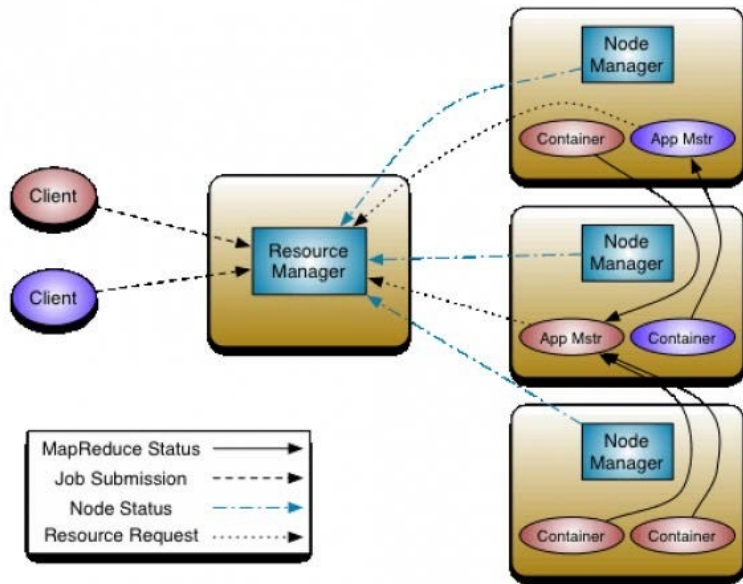
Task Tracker

MapReduce Engine

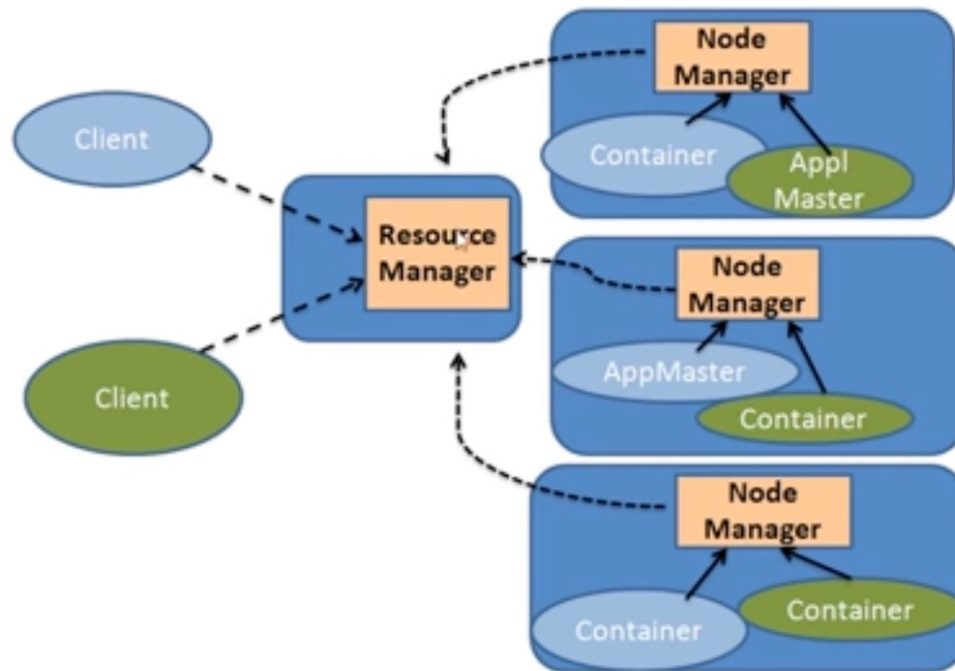


1. JobTracker is a Master daemon
2. Responsible to assign and track task execution progress
3. TaskTrackers are slave daemons
4. They run on systems where data nodes reside
5. Responsible to spawn a child jvm to execute Map, Reduce and intermediate tasks

Apache Hadoop NextGen MapReduce (YARN)



Apache Hadoop NextGen MapReduce (YARN)



✓ Job tracker 1.0 responsibility is now split

- Resource Manager manages the resource allocation in the cluster

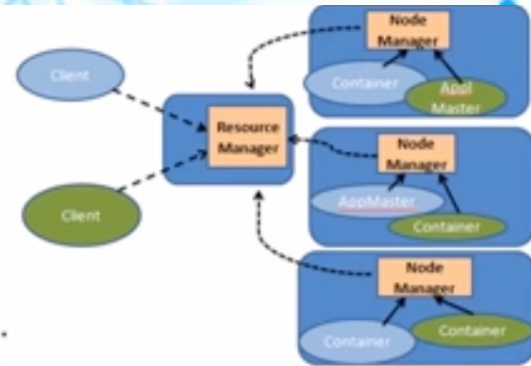
- Application master manages resource needs of individual applications

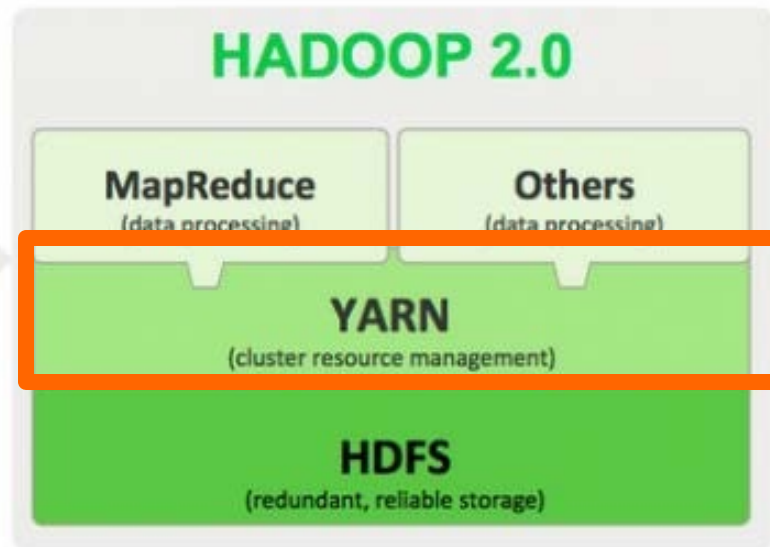
✓ Node Manager is a generalized task tracker

✓ A container executes an application specific process

Apache Hadoop NextGen MapReduce (YARN)

1. **Client:** To submit MapReduce jobs
2. **Resource Manager:** To manage the use of resources across the cluster.
3. **Container:** Name given to a package of resources including RAM, CPU, Network, HDD etc.
4. **Node Manager:** to oversee the containers running on the cluster nodes.
5. **Application Master:** which negotiates with the Resource Manager for resources and runs the application-specific process (Map or Reduce tasks) in those clusters.





Major drawbacks of Hadoop 1.0:

- **Availability** – JobTracker is the only point of availability in Hadoop 1.0. This means that if JobTracker fails, all the tasks will restart by default.
- **Limited scalability** – Since JobTracker is performing multiple tasks and running on a single machine, the other available machines are not being used; hence, resulting in limited [scalability](#).
- **Resource utilization** – In Hadoop 1.0, the map slots and reduce slots are predefined. It might happen that one of the slots is full but the other machine slots are empty. Since the empty slots are reserved, they will sit idle instead of compromising for the full slots. This might cause an issue of resource utilization.
- **Running non-MapReduce applications** – JobTracker is an application which is built for the MapReduce framework. The problem arises when a non-MapReduce application tries to run in this framework. The application needs to conform to the MapReduce framework programming in order to run successfully. Some of the common issues faced due to this include problems with:
 - Ad-hoc query
 - Real-time analysis
 - Message passing approach

- **YARN enhances the power of a Hadoop compute cluster**

Scalability

- **YARN enhances the power of a Hadoop compute cluster**

Scalability

- **Scalability** bottleneck caused by having a single JobTracker. According to Yahoo!, the practical limits of such a design are reached with a cluster of 5,000 nodes and 40,000 tasks running concurrently.
- The computational resources on each slave node are divided by a cluster administrator into a fixed number of map and reduce slots.
- Hadoop was designed to run MapReduce jobs only.

Supports other Workloads

MapReduce Compatibility

Improved cluster utilization

•**Better utilization of resources** – The YARN framework does not have any fixed slots for tasks. It provides a central resource manager which allows you to share multiple applications through a common resource.

•**Running non-MapReduce applications** – In YARN, the scheduling and resource management capabilities are separated from the data processing component. This allows Hadoop to run varied types of applications which do not conform to the programming of the Hadoop framework. Hadoop clusters are now capable of running independent interactive queries and performing better real-time analysis.

•**Backward compatibility** – YARN comes as a [backward-compatible](#) framework, which means any existing job of MapReduce can be executed in Hadoop 2.0.

•**JobTracker no longer exists** – The two major roles of the JobTracker were resource management and job scheduling. With the introduction of the YARN framework these are now segregated into two separate components, namely:

- NodeManager
- ResourceManager

The background of the slide features a complex, abstract network diagram. It consists of numerous small blue dots, representing nodes, which are interconnected by thin, light blue lines. These lines form a web-like structure that spans the entire width of the slide. The nodes and lines are more densely packed in some areas, creating a sense of depth and connectivity. The overall aesthetic is clean and modern, with a focus on the interconnected nature of the network.

The Hadoop “Zoo”

Welcome to the Zoo!



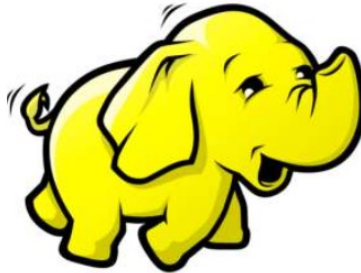
Zookeeper



Pig



Shark



Hadoop

Jaql

Giraph



Hama

I am sure you won't find a Shark in any other zoo ☺



Apache Hadoop Ecosystem



Ambari

Provisioning, Managing and Monitoring Hadoop Clusters



Scoop
Data Exchange



Flume
Log Collector



Zookeeper
Coordination



Oozie
Workflow



Pig
Scripting



Mahout
Machine Learning

R Connectors
Statistics



Hive
SQL Query



Hbase
Columnar Store



YARN Map Reduce v2

Distributed Processing Framework

HDFS

Hadoop Distributed File System

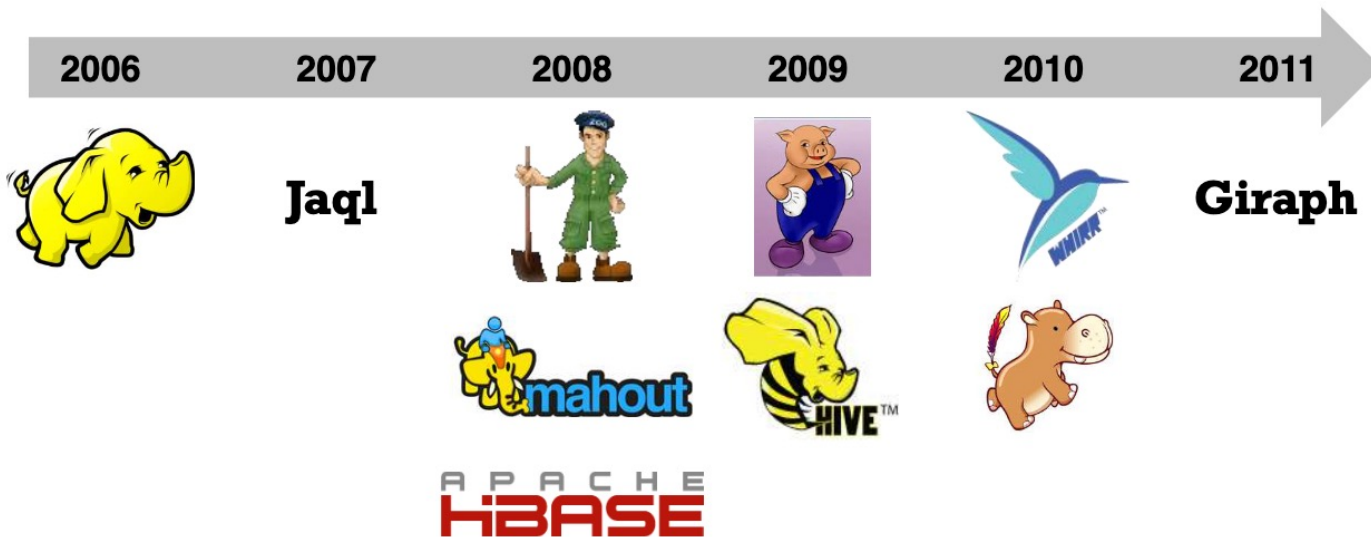


Let's put the 2



Evolution Timeline

- Started by Doug Cutting at Yahoo! in early 2006, and named after his kid's toy elephant
- Hadoop committers work at several different organizations
 - Including Facebook, Yahoo!, LinkedIn, Twitter, Cloudera, Hortonworks

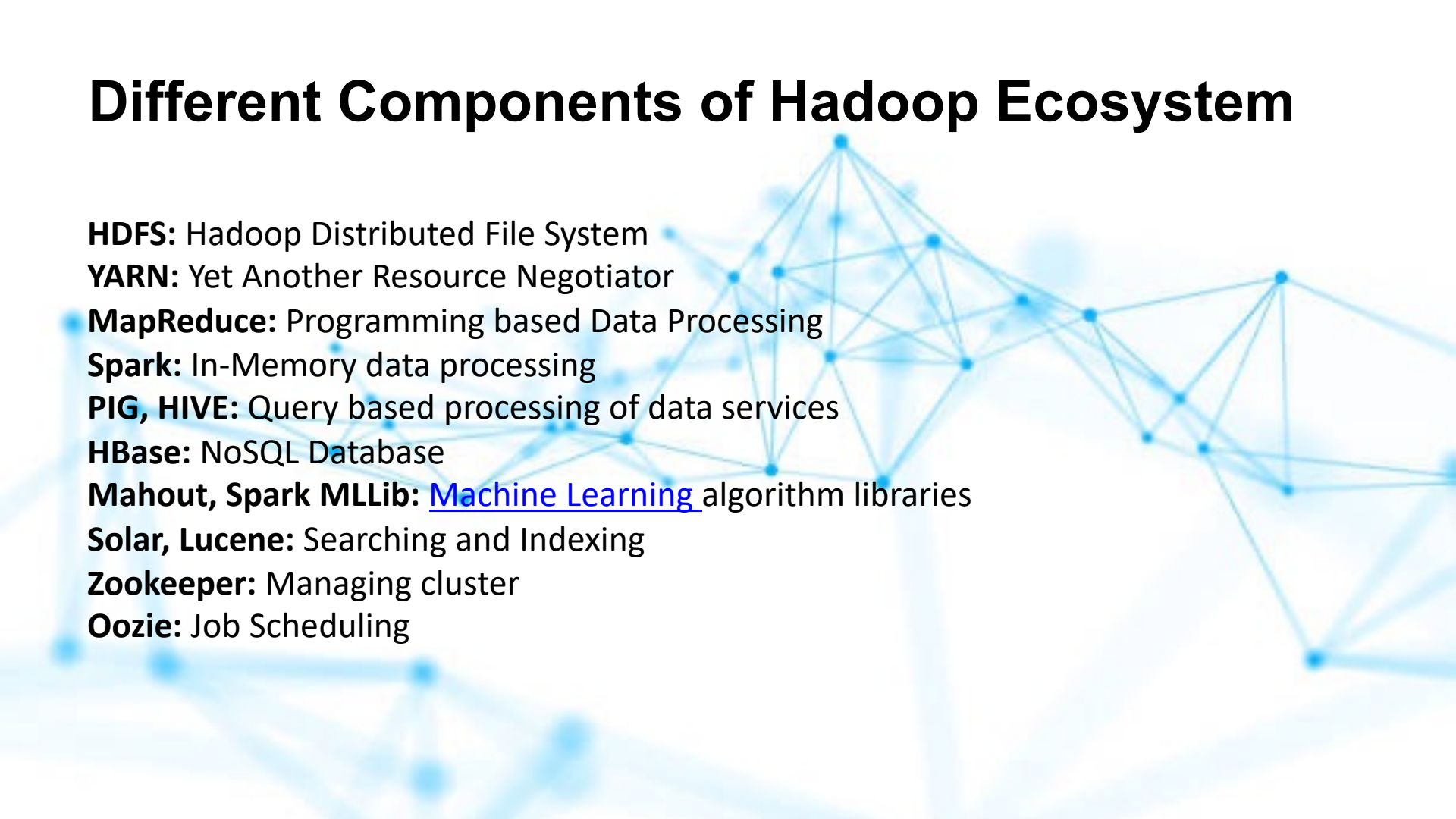


What is Hadoop?

- ❏ Hadoop is an open-source project overseen by the Apache Software Foundation
- ❏ Hadoop is an ecosystem, not a single product
- ❏ Originally based on papers published by Google in 2003 and 2004
- ❏ Some of the projects in the ecosystem have been inspired based on whitepapers published by Google

Google calls it:	Hadoop equivalent
GFS	HDFS
MapReduce	Hadoop MapReduce
Sawzall	Hive, Pig
BigTable	HBase
Chubby	ZooKeeper
Pregel	Giraph

Different Components of Hadoop Ecosystem



HDFS: Hadoop Distributed File System

YARN: Yet Another Resource Negotiator

MapReduce: Programming based Data Processing

Spark: In-Memory data processing

PIG, HIVE: Query based processing of data services

HBase: NoSQL Database

Mahout, Spark MLlib: [Machine Learning](#) algorithm libraries

Solar, Lucene: Searching and Indexing

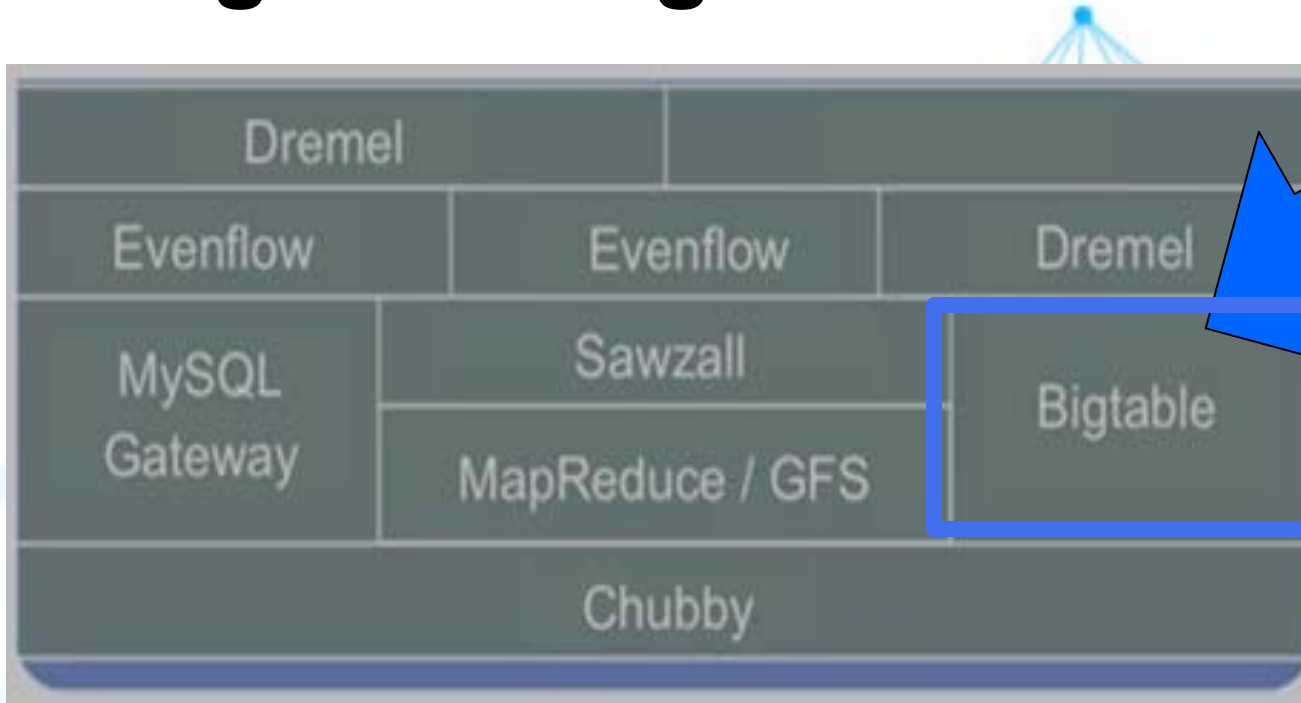
Zookeeper: Managing cluster

Oozie: Job Scheduling

Original Google Stack

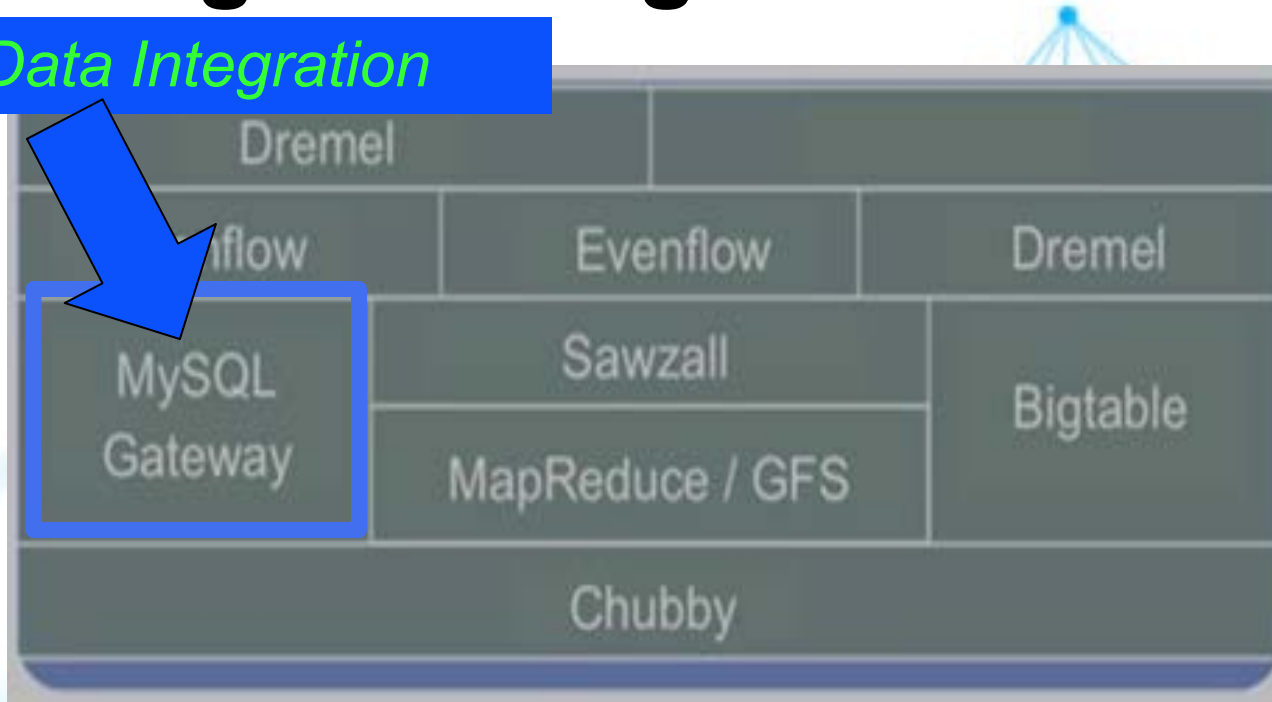


Original Google Stack



Original Google Stack

Data Integration

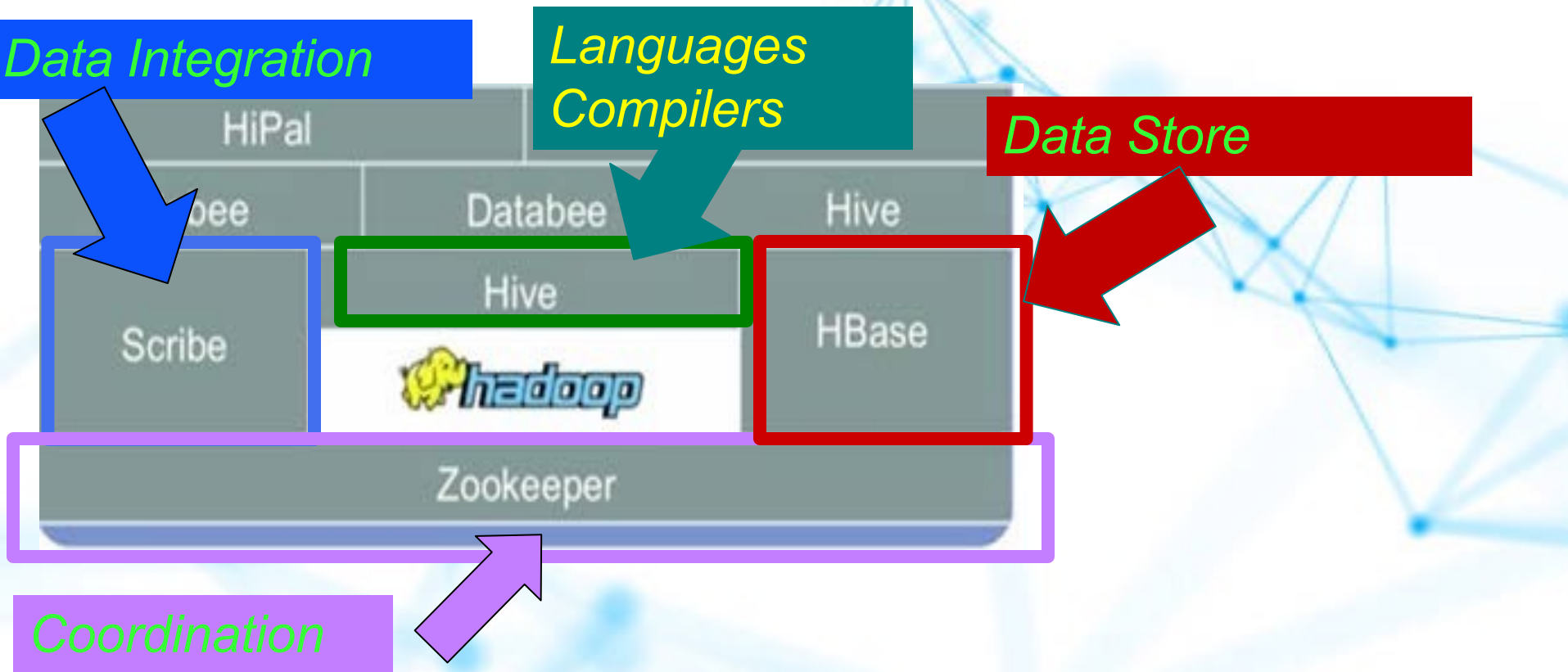


Facebook's Version of the Stack

Data Integration

*Languages
Compilers*

Data Store

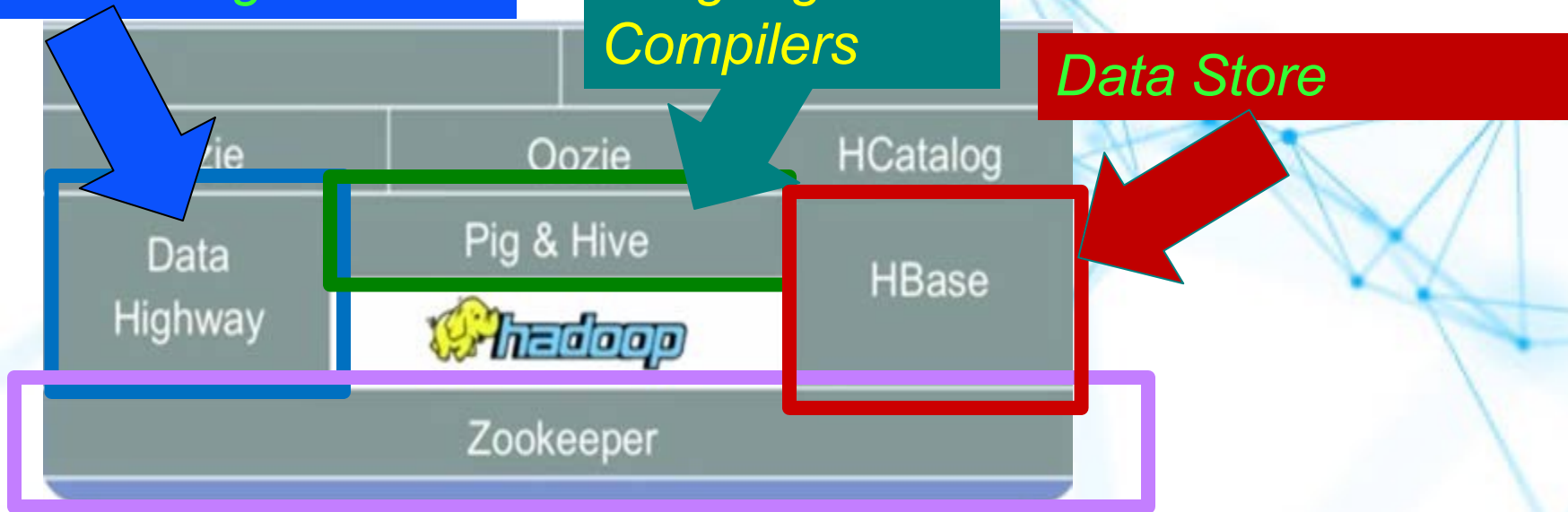


Yahoo's Version of the Stack

Data Integration

*Languages
Compilers*

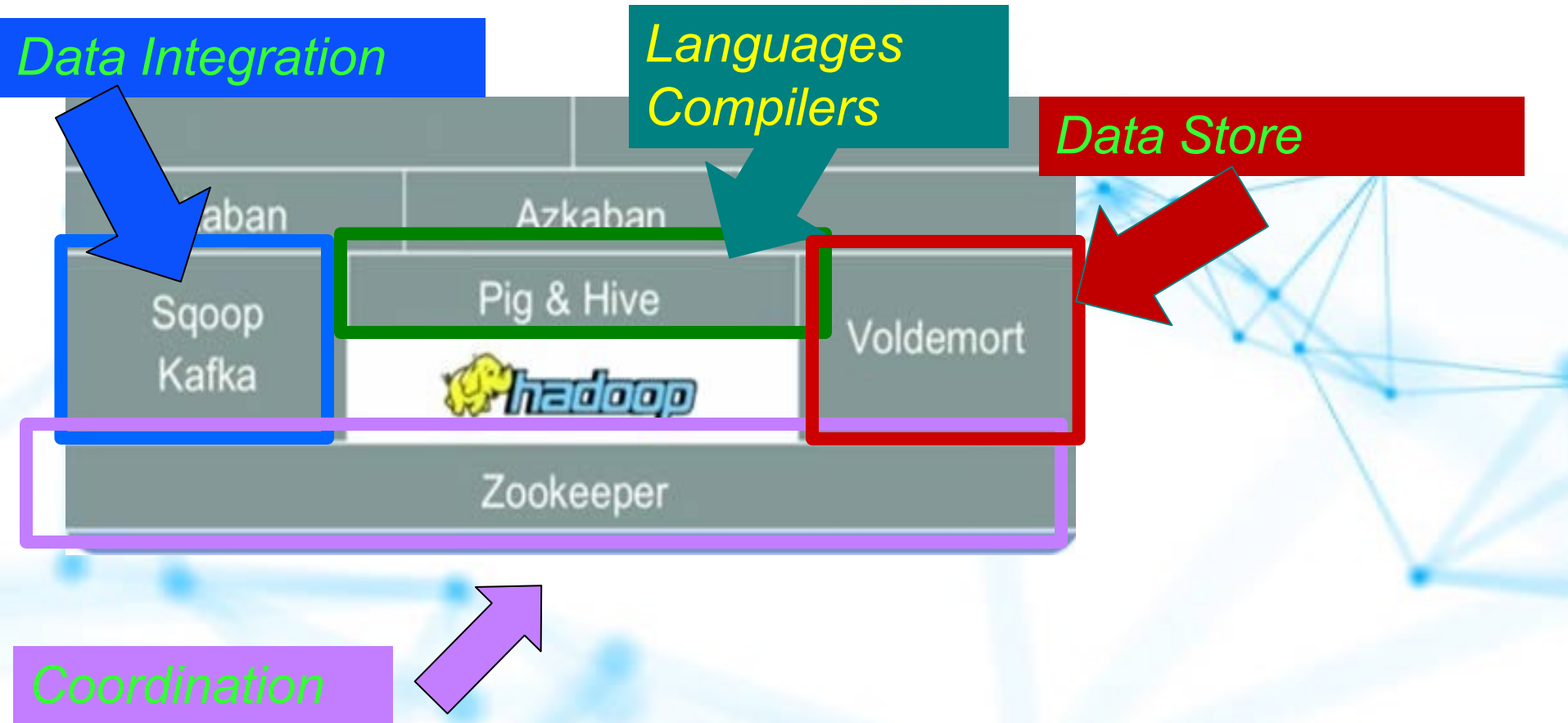
Data Store



Coordination



LinkedIn's Version of the Stack

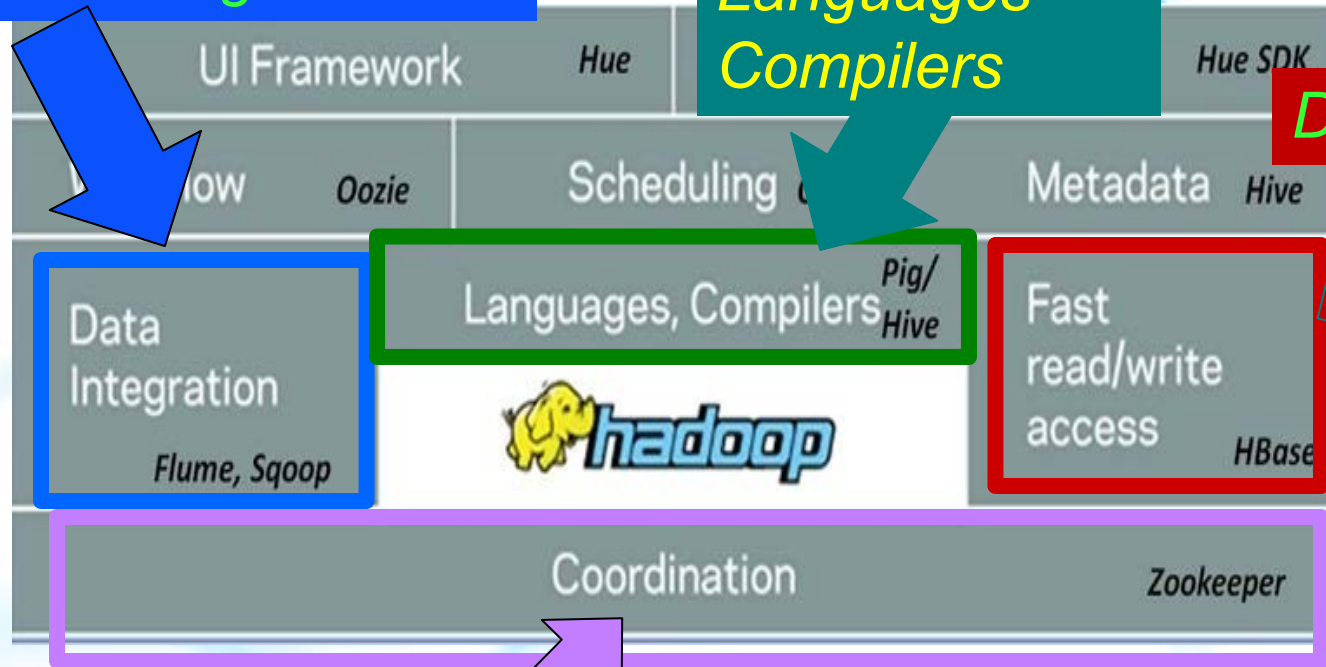


Cloudera's Version of the Stack

Data Integration

*Languages
Compilers*

Data Store



Coordination

The background of the slide features a complex, abstract network diagram. It consists of numerous small blue dots, representing nodes, which are interconnected by thin, light blue lines. These lines form a web-like structure that fills the entire frame, with some areas appearing more densely connected than others. The overall aesthetic is clean and modern, typical of a technical presentation.

Hadoop Ecosystem Major Components

Hadoop Ecosystem

Major Components

- *Hadoop Ecosystem* is a platform or a suite which provides various services to solve the big data problems.
- It includes Apache projects and various commercial tools and solutions. There are *four major elements of Hadoop* i.e. **HDFS, MapReduce, YARN, and Hadoop Common**.
- Most of the tools or solutions are used to supplement or support these major elements. All these tools work collectively to provide services such as absorption, analysis, storage and maintenance of data etc.



Apache Hadoop Ecosystem



Ambari

Provisioning, Managing and Monitoring Hadoop Clusters



Scoop
Data Exchange



Flume
Log Collector



Zookeeper
Coordination



Oozie
Workflow



Pig
Scripting



Mahout
Machine Learning

R Connectors
Statistics



Hive
SQL Query

APACHE HBASE
Columnar Store

Hbase
Columnar Store



YARN Map Reduce v2

Distributed Processing Framework

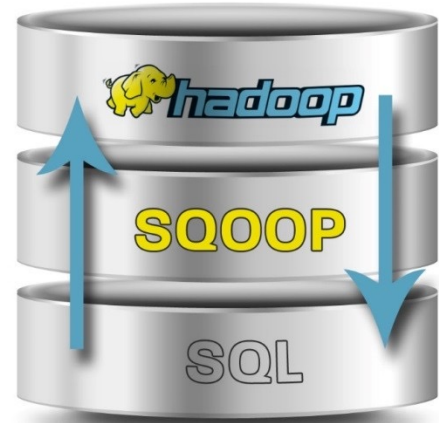
HDFS

Hadoop Distributed File System



Apache Sqoop

- Tool designed for efficiently transferring bulk data between Apache Hadoop and structured datastores such as relational databases





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Hbase
Columnar Store



YARN Map Reduce v2
Distributed Processing Framework

HDFS

Hadoop Distributed File System



HBASE

- Column-oriented database management system
- Key-value store
- Based on Google Big Table
- Can hold extremely large data
- Dynamic data model
- Not a Relational DBMS



Apache Hadoop Ecosystem



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Provisioning, Managing and Monitoring Hadoop Clusters



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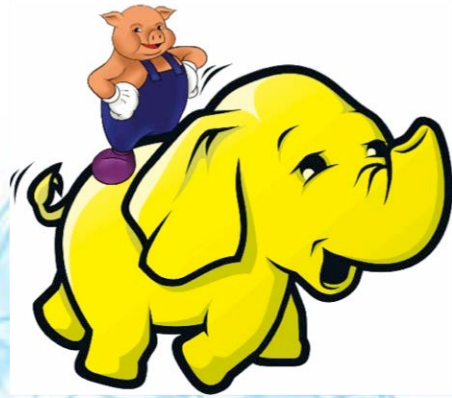
YARN Map Reduce v2
Distributed Processing Framework

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Hadoop Distributed File System

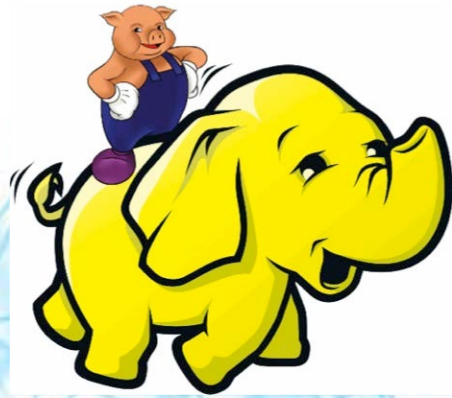


PIG



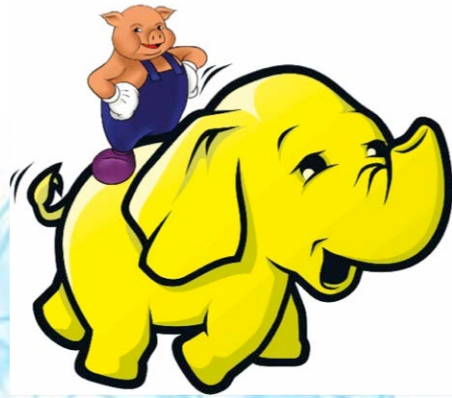
**High level programming on top of
Hadoop MapReduce**

PIG



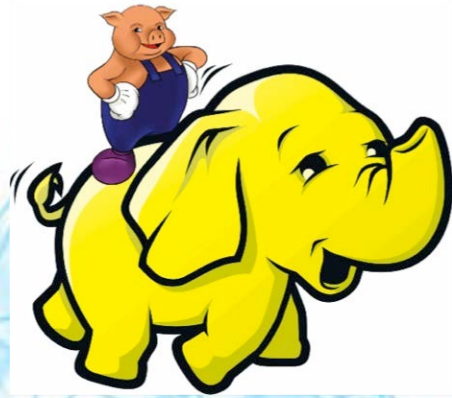
The language: Pig Latin

PIG



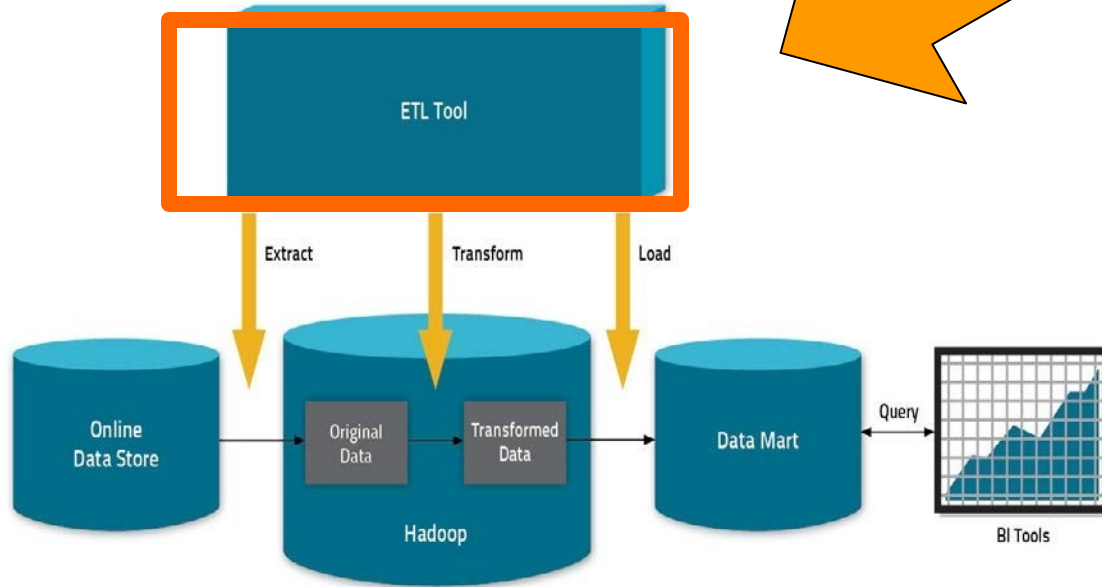
Data analysis problems as data flows

PIG

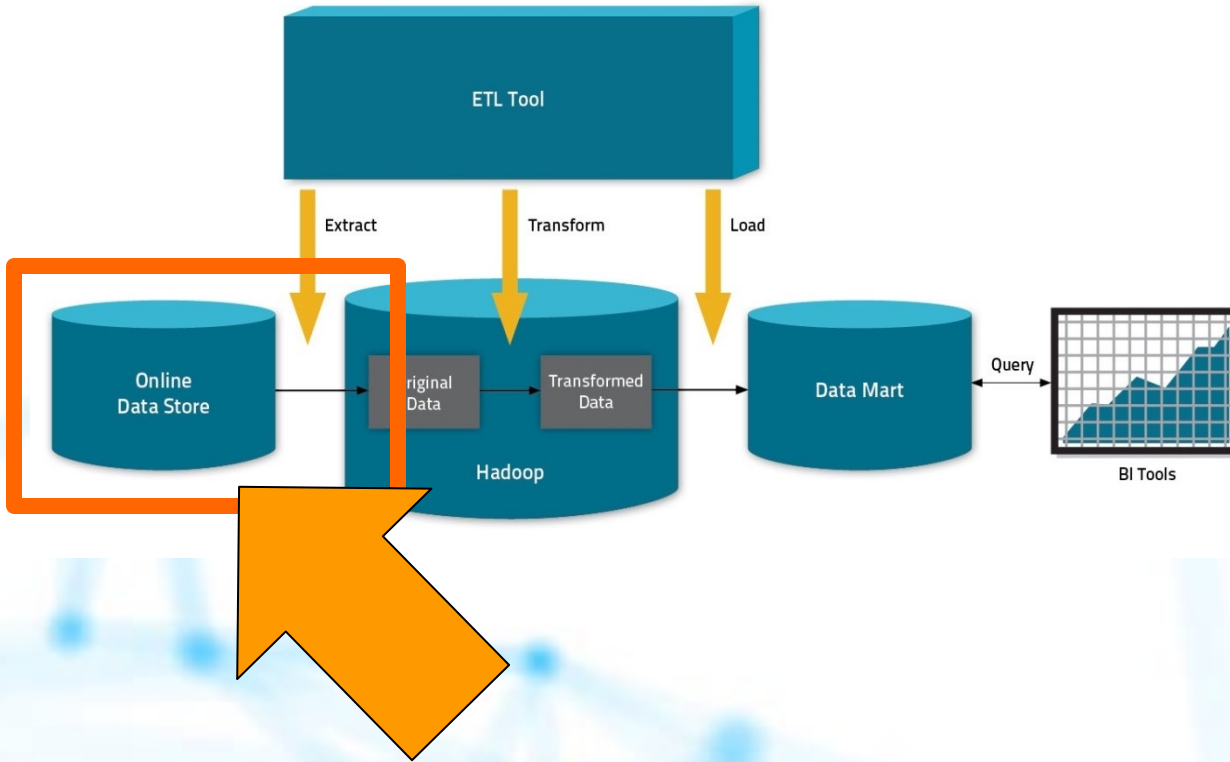


Originally developed at Yahoo 2006

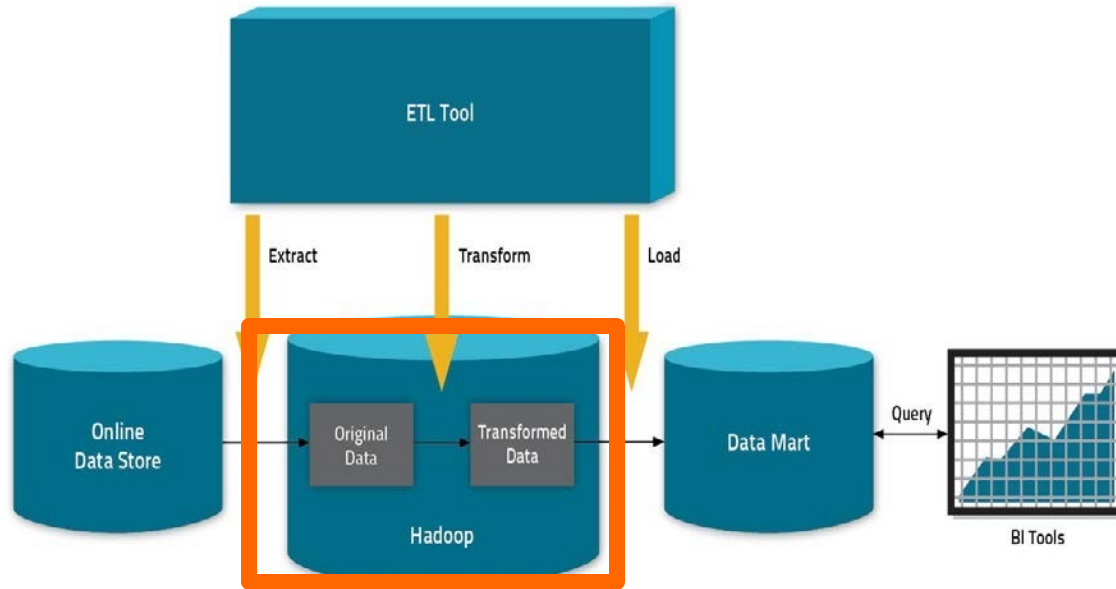
Pig for ETL



Pig for ETL

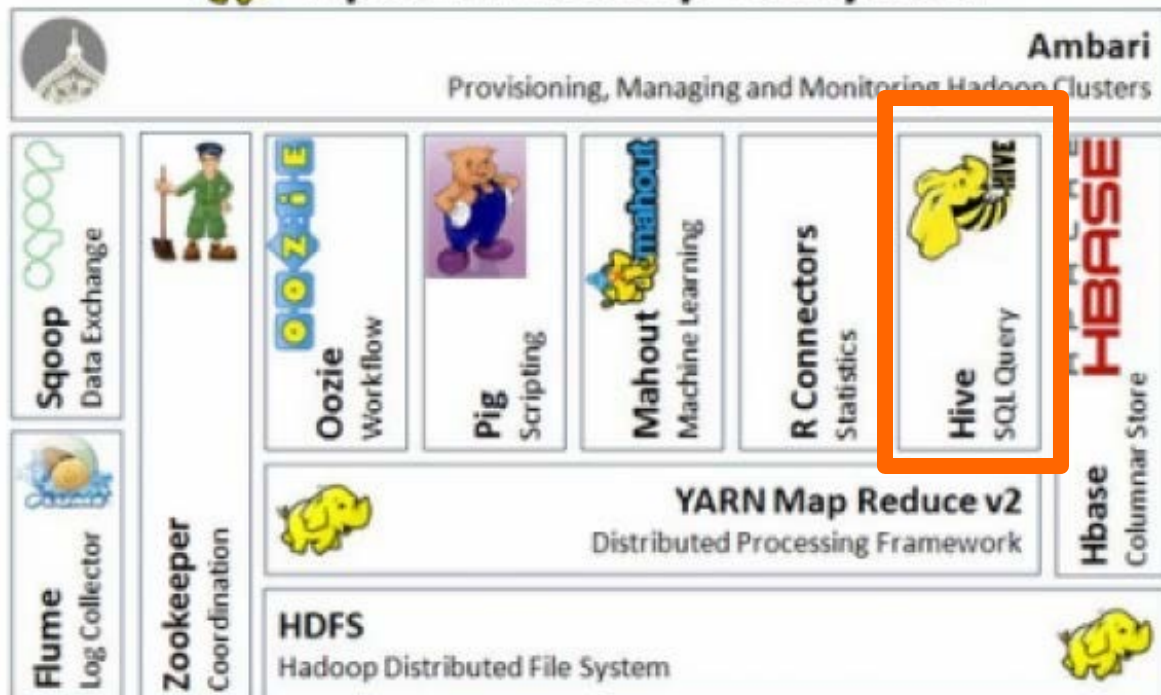


Pig for ETL





Apache Hadoop Ecosystem



Apache Hive



- **Data warehouse software facilitates querying and managing large datasets residing in distributed storage**

Apache Hive

SQL-like language!



Apache Hive

**Facilitates querying and
managing large datasets in
HDFS**



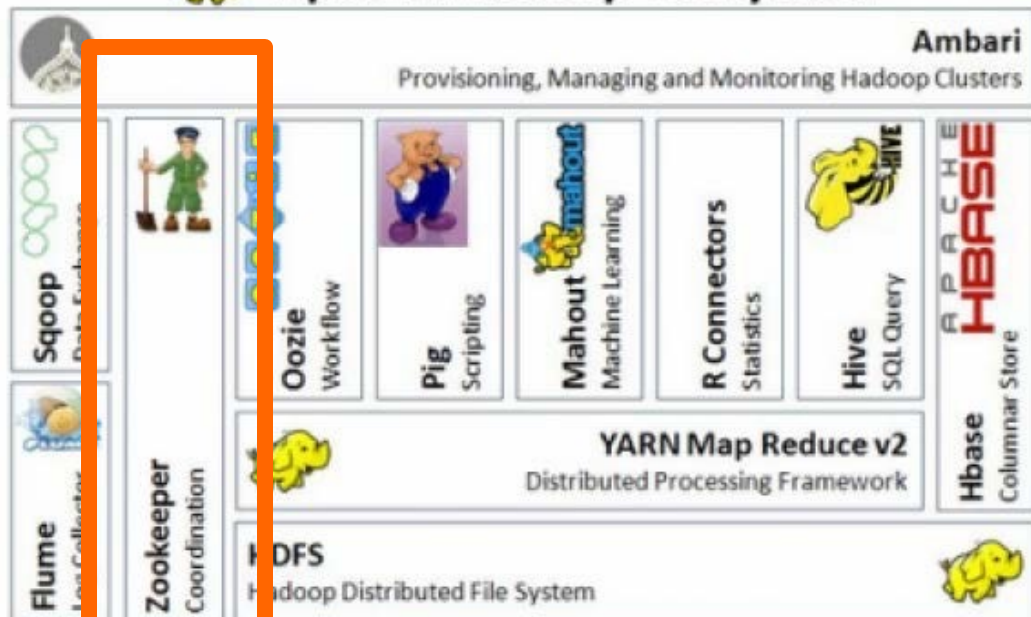
Apache Hive

Mechanism to project structure onto this data and query the data using a SQL-like language called HiveQL





Apache Hadoop Ecosystem



Zookeeper



Provides operational services for a Hadoop cluster group services

Zookeeper

Centralized service for:
maintaining configuration information
naming services
providing distributed synchronization
and providing group services



Flume



Distributed, reliable, and available service for efficiently collecting, aggregating, and moving large amounts of log data